CliffForge

Generative AI for Activity-Cliff-Aware Drug Binding Affinity Prediction

Purpose of this brief

A concise handoff for a teammate explaining the healthcare problem, proposed solution, model architecture, datasets, evaluation plan, and computational requirements. It separates the current concept from implementation choices that still need to be confirmed.

One-sentence idea: Generate small, protein-pocket-aware modifications of known drug-like molecules, estimate how those edits affect binding, and train a pairwise model to predict relative affinity—then test whether this improves predictions on real activity cliffs.

Project status at a glance

Item	Current plan
Healthcare problem	Activity cliffs: similar molecules can have sharply different biological activity / binding affinity.
Core hypothesis	Synthetic, minimally edited ligand pairs may provide useful extra training signal for relative-affinity prediction.
Model family	Pocket-aware molecular generation + scoring/filtering + Siamese/pairwise affinity-difference model.
Hackathon scope	Small proof of concept: ideally one target, a limited congeneric series, and clear baselines.
Important status note	Datasets and software below are candidates from the current proposal, not all finalized selections. The idea is a hypothesis to validate, not a proven improvement.

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1. The problem in plain language

During early drug discovery, researchers often modify a molecule in small ways to improve its interaction with a target protein. Sometimes a tiny chemical edit causes a large change in biological activity. These pairs are called **activity cliffs**. For example, adding or changing a small chemical group can disrupt a hydrogen bond or introduce a steric clash inside the protein binding pocket.

Many predictive models struggle with these cases because they learn broad trends from available data and may treat similar molecules as having similar activity. Activity-cliff examples are relatively sparse, and the key effect can depend on three-dimensional protein–ligand interactions. This can cause a model to rank related candidate molecules incorrectly.

Why healthcare cares

Better early-stage prioritization could help researchers focus experimental effort on more promising compounds. CliffForge is a computational research prototype; it is not a diagnostic tool and does not make treatment decisions.

2. Proposed solution

CliffForge proposes to create additional, challenging molecule-pair examples rather than relying only on naturally available measured pairs. Starting with a known protein–ligand complex, we modify a small part of the ligand while keeping the surrounding protein pocket in view. We then estimate the relative binding performance of the original and modified molecules and use these comparisons to train a model that predicts the difference between two candidates.

Step	Action	What happens
1	**Start with a known complex**	Use a public protein–ligand structure with an associated ligand and, where available, experimental affinity/activity data.
2	**Generate minimal analogues**	Mask or replace a small ligand region using pocket-aware inpainting; use fragment/R-group enumeration as a simpler fallback.
3	**Validate and score**	Check chemical validity and pose plausibility, then compute docking or scoring estimates for both members of each pair.
4	**Train a pairwise model**	Encode each ligand–pocket interaction with shared weights and predict the affinity difference between the two candidates.
5	**Evaluate honestly**	Test on held-out experimental pairs/activity cliffs—not just the synthetic pairs used to train the model.

3. Proposed model architecture

This is the planned pipeline, not a claim that all components have already been implemented.

A. Input
Protein structure + bound ligand pose + ligand chemistry

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B. Local edit
Select a ligand fragment / R-group to mask or replace

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C. Counterfactual analogue generation
DiffSBDD inpainting if feasible; FEgrow / RDKit-based enumeration as a practical fallback

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D. Filtering and labels
Chemical validity + pose checks → Vina/Smina/GNINA scoring → pairwise proxy Δscore

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E. Shared pair encoder (Siamese)
Encode ligand A + protein pocket and ligand B + the same pocket using shared parameters. Candidate representations: molecular graphs and 3D interaction features.

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F. Difference head
Combine the two embeddings and predict Δaffinity / relative ranking. Train with regression and/or pairwise-ranking loss.

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G. Evaluation
Compare against single-molecule and pairwise baselines on held-out experimental data, especially activity-cliff pairs.

Key design choice

The Siamese network uses the same encoder for both molecules, so its learned representations are comparable. The output is a **difference between candidates**, not just two unrelated absolute-affinity predictions. The precise 3D encoder and loss are still implementation decisions to finalize after checking data availability and compute.

Important label caveat

Docking-score differences are noisy proxies, not measured potency. The model must be judged on experimental labels held out from training; otherwise it may simply learn the scoring function's biases. Do not call every generated pair a confirmed activity cliff—call them *counterfactual candidate pairs* until experimental evidence supports the cliff label.

4. Candidate datasets and approximate scale

No single dataset provides all three things we need—3D complex structures, experimental activity/affinity labels, and curated activity-cliff pairs. The likely plan is to combine sources for different roles. Release/version and filtering choices change exact counts; record the version, filters, and final retained counts in the project README.

Dataset	Role in project	Scale / size note	Status
PDBbind	3D protein–ligand complexes with affinity-related data; primary structure source.	Common releases contain thousands of refined-set complexes (roughly 5k in some recent editions) and a larger general set (tens of thousands; roughly 20k in some editions). Exact release must be selected.	Candidate; not finalized
BindingDB	Experimental binding measurements; potential source of real potency/affinity labels.	Millions of reported measurements across targets and assay types; raw count is not the usable count. Requires target, assay, unit, and duplicate filtering.	Candidate; filtering needed
ChEMBL	Curated bioactivity records; help find congeneric series and measured activity cliffs.	Millions of activity records/compound–assay observations in modern releases; usable subset depends heavily on target and assay filters.	Candidate; filtering needed
MoleculeACE	Ligand-only activity-cliff benchmark for evaluating general cliff behavior.	30 target-specific benchmark datasets; per-target sample counts vary. It does not itself provide a standardized structure-based benchmark.	Evaluation candidate
Bajorath 3D-cliff collections / literature-curated pairs	Potential real 3D cliff evaluation subset when ligand structures and target overlap are available.	Published collections report hundreds of 3D-cliff cases across multiple targets; exact usable subset depends on matching structures and affinity measurements.	Optional; verify access
Binding MOAD	Alternative/secondary source of protein–ligand structures and annotations.	Large curated complex collection; usable count depends on release and availability of affinity/activity annotations.	Optional alternative

Recommended hackathon data scope

Start with **one protein target** and a congeneric series for which both structures and experimental measurements can be matched. Build a small, auditable set first. Use synthetic counterfactual pairs only for training augmentation; reserve real experimental pairs for evaluation. MoleculeACE can be a supplementary ligand-only evaluation, but it cannot alone validate the structure-based claim.

Data governance

Use publicly available scientific datasets and molecular structures. Do not use identifiable patient data. Keep a record of dataset licenses, source papers, preprocessing, and train/validation/test splits.

5. Computation and software plan

The current proposal is scoped for a small hackathon proof of concept. The longer research plan previously discussed allows access to approximately 2–3 NVIDIA A100 40 GB GPUs, around 256 GB RAM, and substantial CPU resources through SSH. That is a planning constraint, not confirmation that those resources are available for the hackathon.

Workload	Compute expectation	Practical note
RDKit preprocessing / matched-pair construction	CPU-friendly; memory depends on data size	Deduplicate carefully, standardize units/salts, and keep a reproducible manifest.
Ligand generation / inpainting	GPU recommended; runtime depends on generator and sampling settings	Start with a small number of ligands and 10–50 analogues per parent only if runtime allows. Generation is a likely bottleneck.
Docking / scoring	Usually CPU-parallelizable; GNINA CNN scoring may benefit from GPU	Score parent and analogue under consistent preparation and pocket settings.
Pairwise model training	One GPU is likely enough for a small prototype	Begin with compact graph/3D features, fixed splits, and small batches; scale only after the pipeline works.
Longer-term research scale	Planning target: 2–3 × A100 40 GB, ~256 GB RAM	Could support broader experiments, but does not remove the need for careful scope and storage planning.

Software stack under consideration

Core: Python, PyTorch, PyTorch Geometric, RDKit. **Generation:** DiffSBDD (if pocket inpainting is feasible in the available environment) or FEgrow / RDKit-based fragment enumeration. **Scoring:** AutoDock Vina or Smina; GNINA if environment and GPU permit. **Pose validation:** PoseBusters or PoseCheck. **Optional higher-cost validation:** MM-GBSA or relative free-energy calculations on a very small subset only if setup and compute permit.

Implementation strategy

Build the pipeline in layers: (1) load and standardize one target's complexes; (2) generate/construct analogues; (3) validate and score; (4) train a simple pairwise baseline; (5) add the proposed synthetic augmentation; (6) evaluate on the same held-out experimental pairs. Avoid building a complex 3D architecture before the data and scoring loop works end-to-end.

6. Evaluation plan and success criteria

Experiment	Question answered	Metrics / checks
Baseline comparison	Does synthetic-pair augmentation help at all?	Pairwise ranking accuracy; RMSE/MAE for affinity differences where labels are comparable.
Real-cliff evaluation	Does it generalize to real activity cliffs?	Evaluate only on held-out experimentally measured pairs; report results by target and sample count.
Ablation	Which component contributes?	Pairwise model without synthetic augmentation vs. with augmentation; optionally compare generated vs. enumerated analogues.
Chemistry / pose checks	Are the generated candidates plausible enough to score?	Chemical validity, similarity/edit size, stereochemistry checks, clashes/pose plausibility, and duplicate rate.
Leakage controls	Are results inflated by near-duplicates or shared series?	Split by ligand/scaffold/series where feasible; remove duplicates and ensure test pairs are unseen.

7. Main risks and how we handle them

Noisy proxy labels: docking scores do not equal experimental affinity. Treat them as weak supervision and test on real data.

False or artificial cliffs: a generated pair with a large score difference is not automatically a real activity cliff. Use cautious naming and experimental evaluation.

Bad generated poses: apply chemistry and pose checks before scoring.

Prior-work overlap: pairwise affinity models and negative/decoy augmentation already exist. The proposed contribution is specifically to test pocket-aware minimal-edit counterfactual augmentation for activity-cliff sensitivity, not to claim the Siamese architecture itself is new.

Scope risk: start with one target and a strong ablation. Do not promise an unlimited supply of reliable labels or a definitive new benchmark until quality and coverage are demonstrated.

8. Suggested division of work

Workstream	Tasks
Data	Choose target, obtain public structures/affinities, standardize compounds, construct leakage-aware splits.
Generation + scoring	Set up inpainting or enumeration, prepare ligands/protein, validate poses, run consistent scoring.
Model	Implement pairwise baseline and synthetic-augmentation variant; log seeds and configs.
Evaluation + reporting	Run held-out tests and ablations; document failures, data versions, source acknowledgements, and limitations.

9. Starter references and tools to verify

These are starting points for the team to verify and cite in the final submission. Confirm the exact paper version, release, license, and implementation details before relying on specific numbers.

• van Tilborg, Alenicheva & Grisoni (2022), “MoleculeACE: a benchmark for evaluating molecular machine learning on activity cliffs,” Journal of Chemical Information and Modeling.

• PDBbind database and its release-specific documentation (protein–ligand structures and affinity annotations).

• BindingDB and ChEMBL documentation (experimental binding/activity measurements and assay metadata).

• DiffSBDD documentation/paper (structure-based molecular generation; verify whether the chosen checkpoint supports the required inpainting workflow).

• AutoDock Vina / Smina / GNINA documentation for scoring and docking; PoseBusters or PoseCheck for structure plausibility checks.

Bottom line for the teammate: We are not claiming that we have already solved activity-cliff prediction. We are testing whether generating pocket-aware, minimally edited molecular pairs and training a pairwise affinity model on approximate relative scores can improve performance on real, unseen activity cliffs. The strongest proof is a clean comparison against baselines on held-out experimental data.